

Lectures 9 & 10

In Class Exercises



Exercise 1

- Using the figure in the slides as a model, illustrate the operation of COUNTING-SORT on the array $A = \{6, 0, 2, 0, 1, 3, 4, 6, 1, 3, 2\}$.

Exercise 2

- For the set $\{1, 4, 5, 10, 16, 17, 21\}$ of keys, draw binary search trees of heights 2, 3, 4, 5 and 6.

Exercise 3

- Give a recursive algorithm that performs preorder and postorder tree walks in $\Theta(n)$ time on a tree of n nodes.

Exercise 4

- Write recursive versions of TREE-MINIMUM and TREE-MAXIMUM.

Exercise 5

- Write the TREE-PREDECESSOR procedure.